

HCI+2026 | Human-Centered Intelligence+暑期研究项目报名开始+项目信息

HCIX Research Club 2026年5月7日 21:57 北京

项目介绍

Human-Centered Intelligence+ (HCI+) 是由清华大学、香港城市大学、香港科技大学（广州）等多所国内外知名高校和机构的人机交互领域知名教授联合组织操办。此项目结合了往年广受欢迎的暑研项目“**HCI·X**”与“**Access Computing Summer Program (ACSP)**”的优质资源和经验。我们旨在为全球热爱人机交互与智能系统的优秀学子搭建一个接触和学习人机交互研究的多元、开放、协作的平台。

项目聚焦以人为本的智能技术创新，鼓励多学科背景的学生通过团队协作和实践项目，探索如何利用人工智能和人机交互推动社会福祉、健康、无障碍、可持续发展等重要议题。参与者将在顶尖导师团队的带领下，深入体验学术前沿，收获宝贵的研究与创新实践经历。

每位参与者将以小组形式，加入由一名导师带领的具体应用研究项目，在团队合作中深入体验真实的科研与创新过程。项目期间还将特别邀请来自学术界与业界的人机交互、人工智能等领域的知名专家开设系列专题讲座和前沿课程，系统拓展同学们的人机交互视野，提升跨领域研究能力。

2026年，HCI+ 暑期项目将招募若干远程工作的实习生参与到各位导师的多个人机交互课题研究中。申请者将参与到项目中至少三个月，每周至少30个小时。项目持续录取，各项目导师将会于5月开始面试，申请结果将由导师团队综合决定。

我们相信，真正有温度的智能，源于**对“人”的理解**。欢迎热爱创新、富有同理心与社会责任感你，加入HCI+，与全球志同道合的伙伴一起共创未来！

- 项目时间：2026年6-9月（具体由各项目导师自定）
- 地点：**远程线上**
- 申请截止日期：**滚动制录取，最晚于5月20日及之前提交申请**
- 报名方式：在线申请表（见文末）

HCI+ 导师团队

HCI+2026 | Human-Centered Intelligence+暑期研究项目导师介绍

具体项目内容

（按导师姓名首字母A~Z排序，排名不分先后，具体项目要求请登录报名系统查看）



Pengcheng An
Assistant Professor,
Southern University of Science and Technology



01. Multimodal and Agentic AI for Children's Playful Learning

This project explores how multimodal and agentic AI systems can support children's learning in specific educational tasks through playful, creative, and interactive experiences. Students are encouraged to move beyond existing learning tools and interaction paradigms, and to imagine new forms of child-AI interaction involving cameras, speech, vision, multi-agent collaboration, tangible interfaces, or embodied devices. The project welcomes both software-oriented prototypes and hardware-integrated explorations, especially those that rethink how AI can perceive, respond to, scaffold, and participate in children's learning activities.



Xinyi Fu

Associate Professor

Tsinghua University



01. Immersive AI-Augmented Interaction Systems for Expansion of Human Emotion and Perception

This project explores the development of immersive, AI-augmented interaction systems designed to expand the boundaries of human emotion and perceptual dimensions. By integrating multimodal sensory feedback with real-time generative intelligence, we aim to construct environments that transcend traditional interaction constraints, enabling users to perceive, express, and regulate complex emotions in unprecedented ways. This research focuses on both perceptual augmentation and the creation of novel psychological spaces within human-AI symbiosis, providing critical theoretical frameworks and design insights for the future of immersive affective computing and cognitive expansion.

Required skills

LLM based multi-agent systemA solid academic foundation in Design, Computer Science, Human-Computer Interaction, Informatics, or Psychology. Proficiency in developing immersive environments using Unity or Unreal Engine. Experience in building and deploying applications for XR hardware. Ability to create low-to-high fidelity prototypes and design intuitive user interfaces within 3D/spatial environments. A strong self-motivated mindset with the ability to take ownership of research tasks and drive them from conception to evaluation. Rich experience in simulation games; Game Development Basics; Unity Development; C# programming

Preferred skills

Experience leveraging Large Language Models (LLMs) or generative AI APIs to create intelligent, responsive agents within immersive scenes. Knowledge of integrating biometric sensors, eye-tracking, or spatial audio to enhance human perception and emotional engagement. A track record of contributing to research papers in top-tier venues such as CHI, CSCW, UIST, SIGGRAPH, ISMAR, or IEEE VR, etc. Experience with shader programming, 3D modeling, or animation to create high-quality, emotionally resonant visual environments.



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Nan Gao

Associate Professor

Nankai University



01. AI-Mediated Real-Time Intervention for Parent-Child Communication

Parent-child conflict during everyday interactions often stems from communication breakdowns that go unaddressed in the moment. This project designs and evaluates an LLM-powered system that monitors conversational dynamics in real time and delivers context-sensitive, in-situ nudges to help parents and children de-escalate tension, foster mutual understanding, and build healthier long-term relationships.

Required skills

Background in Human-Computer Interaction, Psychology, Education, or related fields; experience in qualitative or mixed-methods user research; basic understanding of conversational AI or NLP concepts

Preferred skills

Experience with LLM-based application development (e.g., prompt engineering, API integration); familiarity with family studies, child development, or behavioral intervention design; prior work involving sensitive user populations or field studies



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Jiangtao Gong

Associate Professor

Shanghai Jiao Tong University

01. Brain-Inspired Cognitive Companion Agents

This project focuses on building brain-inspired cognitive companion agents by modeling human cognitive patterns in complex cognitive activities using neural data. We aim to better understand how humans perceive, attend, decide, and collaborate in dynamic tasks, and to explore effective intervention methods that can improve human cognition and performance. Based on these insights, the project will further establish benchmarking frameworks for human-AI collaboration, supporting the development of intelligent systems that can work with humans in a more adaptive, interpretable, and cognitively aligned way.

Required skills

Background in Cognitive Science, Human-Computer Interaction, Artificial Intelligence, or related fields; basic understanding of human cognition, experimental design, and data analysis; interest in human-AI collaboration, neural data modeling, and cognitively aligned intelligent systems

Preferred skills

Experience with EEG, eye-tracking, or other physiological sensing technologies; familiarity with behavioral experiments, signal processing, machine learning, or computational cognitive modeling; experience in benchmarking, human-subject studies, or intervention design



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Feilin Han

Assistant professor

Tongji University



01. Who Directs the AI? Narrative Agency in AI-Generated Video

As generative AI moves toward long-form storytelling, the distribution of creative control remains unresolved. This project investigates Narrative Agency—the power to shape the story—across three distinct models: Director-Driven (manual), User-Driven (interactive), and AI-Driven (automatic). This project will explore how these agency structures influence Watchability and Engagement without causing Narrative Fatigue. The findings will provide a design framework for future generative tools, ensuring they support meaningful human authorship while leveraging AI’s creative potential.

Required skills

Strong interest in AIGC and human-AI interaction. Experience with generative video frameworks and the ability to build pipelines with visual consistency. Familiarity with cinematic quality assessment and user study. Proficiency in qualitative and quantitative data analysis.

Preferred skills

Background in film studies and creative storytelling. Basics know-how or interest in computer vision methods. Experience in data collection methods (e.g., semi-structured interviews, questionnaires, focus group).



Yang Jiao

Assistant Professor

Tsinghua University



01. AI Smart Glasses Enabled Accessible Voice Navigation System for Visually Impaired Users

This project explores AI-powered smart glasses that provide accessible voice navigation for visually impaired users. It develops a computer vision system that interprets surroundings in real time—detecting obstacles, signs, crosswalks, and points of interest—and converts them into concise, context-aware voice guidance.

The system will be evaluated against real-person video call assistance through controlled experiments in indoor and outdoor settings, measuring recognition accuracy, response time, user trust, cognitive load, and adaptability. The study also addresses current AI limitations, such as misidentification in complex scenes and weak social context understanding. Proposed improvements include integrating multimodal sensors, enabling continual learning, customizing voice output, adding uncertainty prompts, and creating a hybrid human-AI collaboration model.

Required skills

Background in Design, Human-Computer Interaction, user study and system evaluation

Preferred skills

Experience with Artificial Intelligence, Computer Vision, and Vibe coding ability

Guanhong Liu

Associate Professor

Tongji University

**01. MorphLearn: Designing Interactive Science Learning Toolkits with Shape-Changing Materials**

MorphLearn aims to make abstract scientific concepts more tangible by transforming advanced shape-changing materials into an interactive science learning toolkit for children and teenagers. Through material transformations such as expansion, compression, bending, and volumetric morphing, the project explores how learners can understand natural science mechanisms through hands-on physical interaction. Building on the team's expertise in HCI and smart morphing materials, including LaserMorph, the project focuses on three directions: developing modular toolkits that lower the barrier to creating shape-changing objects; using generative AI to guide users from creative ideas to feasible physical parameters and scientific explanations; and evaluating how the toolkit supports scientific inquiry, creative reasoning, and systems thinking.

Required skills

UI/UX Design, prototyping, or digital fabrication (laser cutting/3D printing);
Basic programming (Arduino/ESP32 or frontend); Literature review and user study design; Visual communication or video production

Preferred skills

Hands-on prototyping and collaborative iteration experience; LLM API/prompt engineering or generative AI tools experience; Background in EdTech, material design, or curriculum development

RAY LC

Assistant Professor

City University of Hong Kong



01. Leveraging AI Identities and Psychological Affordances for Multi-Tool Creative Workflows

Our previous published work at CHI 2026 shows how machine humor can be given a human touch when giving machine agents an identity of the machine itself. How do we optimize creative workflows that involve use of different tools for each step of the creative process by harnessing the tools' own peculiarities? This project uses LLM models to represent each creativity tool in a multi-step workflow. It directs users to think about what the tools themselves are good for based on their own advantages, limitations, and identities.

Required skills

Two of the following items: qualitative / quantitative data analysis, HCI user testing experience, logical and empirical and concise English academic writing, excellent communication / organization / time management skills, graphic design in prototype creation and scientific communication, development skills in LLMs and chatbots and web-based applications.

Preferred skills

Demonstrated motivation and commitment to a CHI submission based on previous evidence and current availability to contribute to our highly competitive research endeavor. Project direction is already set for rapid execution, but student inputs can help tune the project for CHI.



Jiannan Li

Assistant Professor

Singapore Management University



01. Generative Learning Materials for Math Education

This project will explore the student and teacher experiences with AI-generated interactive learning materials for math.

Required skills

AI for education/Agent programming/Quant data analysis (one of them)



Yue Li

Associate Professor

Xi'an Jiaotong-Liverpool University

01. Multi-Agent Scaffolding for Embodied Literacy in Virtual Cultural Environments

This research explores the design of a multi-agent system framework aimed at scaffolding student learning within virtual cultural environments. While 3D reconstructions facilitate the sense of presence, they often lack the pedagogical structure necessary to foster deep historical literacy. This project investigates how specialized AI agents, acting as curators, historical witnesses, and peer-mentors, can provide adaptive support to learners as they navigate complex cultural narratives. By integrating embodied intelligent agents into a virtual environment, the study seeks to transform passive observation into active inquiry-based learning and validate a prototyped workflow that balances historical accuracy with interactive edutainment.

Required skills

Students with strong motivation, good academic records, sound technical skills, excellent communications skills and teamwork ability are desired.

Preferred skills

Specific skills involved in this project include

- Multi-agent framework development & prompt engineering
- Academic paper reading, summary, and sharing
- User study design and implementation



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Dong Ma

Associate Professor

University of Cambridge



01. Vibration-Based User Identification and Authentication via Everyday Input Device

This research explores the use of micro-vibration patterns generated during everyday mouse interactions as a passive signal for user identification and continuous authentication. Unlike traditional biometrics that require explicit input, vibration signals induced by hand movement, grip dynamics, and interaction habits offer an unobtrusive and continuous alternative for recognizing users in situ. This project investigates whether such fine-grained signals can be reliably captured using commodity sensing setups and modeled to distinguish between individuals across sessions. By integrating signal processing with machine learning, the study aims to develop a lightweight pipeline that supports both identity recognition and adaptive personalization, enabling systems to dynamically adjust interfaces, preferences, and security levels based on the current user. Ultimately, this work seeks to demonstrate how interaction-driven sensing can enable seamless, privacy-aware user modeling in everyday computing environments.



Xin Tong

Assistant Professor

Hong Kong University of Science and Technology
(Guangzhou)



01. Social Interaction Assistance Software for Visually Impaired Users

This project explores software-based social assistance for visually impaired individuals. It develops a real-time facial recognition and auditory feedback system that helps users identify approaching acquaintances and respond proactively, reducing social awkwardness. The system provides voice prompts such as "Someone resembling [Name] is approaching," enabling natural social interaction. It also examines how technology can compensate for the lack of public etiquette norms around self-identification, promoting inclusive communication and social participation.

Required skills

Computer vision or software engineering background; Python proficiency (face recognition, real-time inference); Mobile or cross-platform development experience; Familiarity with AI agent frameworks

Preferred skills

Frontend development (React/Vue); UI/UX and accessibility design; Cloud deployment and API integration; Understanding of social computing



Qi Wang

Associate Professor

Tongji University



01. XR-based Cognitive-Motor Training for Ageing Population

This project explores the design and development of extended reality (XR) systems—spanning VR, AR, and MR—to support cognitive-motor training for aging populations. Grounded in principles from rehabilitation science, human-computer interaction, and immersive technology, students will investigate how XR environments can simultaneously engage older adults in physical movement and cognitive exercises to promote healthy aging and functional independence. The project involves building interactive XR prototypes, integrating motion-tracking or sensor-based input, and iteratively refining the system through user feedback. Students will have the opportunity to explore how immersive media can be tailored to the physical and cognitive characteristics of older users.

Required skills

Required skills Proficiency in Unity development (C#) for XR applications, or hands-on experience with hardware prototyping (e.g., microcontrollers, motion sensors, or wearable devices); good understanding of human-centered design principles.

Preferred skills

Preferred skills Familiarity with XR platforms and SDKs (e.g., Meta Quest, HoloLens); interest in or prior exposure to aging, rehabilitation, or health-related technologies; experience with user research methods or HCI will be helpful; knowledge of real-time interaction systems or sensor data handling is a plus.



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Yun Wang

Associate Professor

School of New Media Art and Design, Beihang University



01. Adaptability or Predictability: Generative Strategies for Enhancing Perceived Safety in Human-Robot Arm Collaboration

Should your robot coworker be smarter or more predictable? This study investigates how generative models should balance adaptability and predictability to enhance humans' perceived safety during robotic arm collaboration. It develops a multimodal generative system that evaluates human states in real time and produces adaptive explanations and haptic feedbacks. Through user experiments, it examines the effectiveness of different generative strategies on perceived safety and trust calibration.

Required skills

Background in HCI, Design, CS or Psychology; Experience in conducting user studies, experiments and data analysis; Strong self-motivation and a collaborative team spirit.

Preferred skills

Experience with LLM-based system development or robotic platform programming



Yuntao Wang

Associate Professor

Tsinghua University



01. Egocentric Multimodal Intention Understanding with Smart Glasses

This project aims to build smart-glasses AI assistants that can understand users' intentions by integrating egocentric video, audio, gaze, and wearable signals. Based on a newly collected egocentric dataset, students will develop and evaluate multimodal models for user-centered tasks such as requests, memory, emotional experience, and personal preferences in everyday scenarios.

Required skills

Python, PyTorch, LLM/AI Agent development, multimodal data processing, machine learning fundamentals, scientific reasoning, reproducibility, strong self-motivation

Preferred skills

Vision-language models, egocentric video understanding, wearable sensing, gaze or physiological signal processing, user study, dataset/benchmark construction



Shirley Xue

Assistant Professor

Purdue University



01. Cross-Site Photoplethysmography for Reliable Pulse Rate Variability Monitoring in Daily Life

This project investigates how PPG signals from different wearable body sites can enable reliable, continuous heart rate variability (HRV) monitoring in daily life. HRV is an important physiological marker of autonomic nervous system regulation and is closely associated with stress, recovery, sleep, and mental wellbeing. However, current wearables typically provide only intermittent HRV measurements, often under still or low-motion conditions. Estimating HRV continuously during daily life remains challenging because PPG signals are highly sensitive to motion artifacts, sensor placement, and contact quality. Using a time-aligned dataset of earring, ring, and watch PPG with ECG-derived HRV as ground truth, this project will evaluate state-of-the-art PPG-based HRV estimation methods across body sites, wavelengths, and time windows. It will further explore whether cross-site PPG modeling and sensor fusion can improve HRV reliability, and how continuous HRV patterns relate to self-reported daily activities such as rest, walking, eating, work, and exercise.

Yukang Yan

Assistant Professor

University of Rochester



01. Enhancing User Awareness in AR Social Contexts

This project investigates how users perceive and act in everyday social situations, such as interacting with others or coordinating shared tasks. Using egocentric multimodal data (e.g., first-person video, motion, and gaze), the student will analyze user behavior and develop models to better understand user attention and actions. The goal is to design approaches that help users build greater awareness and understanding of their own behavior in social contexts.

Required skills

Strong programming skills; Background in machine learning and HCI; Experience with data analysis or experimental data; Familiarity with AR/VR

Preferred skills

Experience with user studies or behavioral data collection; Interest in social computing or interaction design

Xin Yi

Associate Professor

Tsinghua University

01. Usable Privacy for Multi-Modal Agentic Interaction Techniques

Agentic interaction techniques have become increasingly popular on multi-modal devices, including GUI, smartphones, AI glasses, etc. However, the functionality of these techniques relies heavily on pervasive sensing and interaction context, which poses severe privacy concerns. This project explores how users and agents can detect and effectively control privacy through collaboration. Specifically, we will consider scenarios including AI glass interaction, embodied interaction, chatbot, etc.

Required skills

Agentic Interaction, User Study, Design, Basic understanding of LLM algorithms

Preferred skills

Frontend development, privacy-related experience, computer vision or VLM experience



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Chun Yu

Associate Professor

Tsinghua University

01. Self-Evolving Personal AI Assistant via Human-AI Co-Evolutionary Interactive Learning

This project develops a self-evolving personal AI assistant empowered by modern generative AI and agentic technologies. Drawing on harness engineering methodology, it builds human-AI interactive learning and co-evolution mechanisms, allowing the assistant to continuously upgrade its intelligence and service capabilities through daily user interaction, behavioral adaptation and real-world usage feedback. The project involves the design of assistant functions that support users in various scenarios—including digital information environments (such as PC and mobile devices), social interactions, and physical world communications (e.g., face-to-face discussions and on-site assistance). The specific scenario and functional direction of the assistant will be fully discussed and determined together with the student, as we highly value and fully respect the student's own interests, believing that intrinsic interest is the core driving force for high-quality research and innovation.

Required skills

Basic AI-augmented programming and rapid prototyping ability; fundamental UI/UX and interaction design literacy; core understanding of human-computer interaction and AI agent systems.

Preferred skills

Strong creativity, independent thinking, outstanding execution ability and bold imagination; recognition that AI will reshape human-computer interaction and drive exponential growth of collective intelligence with profound societal and personal impact; self-motivated to explore research directions based on personal interest; no prior experience in academic paper writing is required, as professional guidance will be fully provided throughout the paper preparation and submission process.

Hao Zhang

Associate Professor

Institute of Software Chinese Academy of Sciences

01. Designing Imperfect AI Companions for Children's Exploratory Play

This project investigates the design of an "imperfect AI companion" to support exploratory play and open-ended learning for young children and neurodiverse populations, including individuals with Autism Spectrum Disorder. Unlike conventional AI systems that prioritize accuracy and task completion, this research proposes a novel paradigm in which controlled imperfection—such as intentional, recoverable errors and uncertainty—is leveraged as a resource to stimulate curiosity, reflection, and active engagement.

Required skills

Background in machine learning, natural language processing, or interactive AI systems; Knowledge of human-computer interaction and user-centered design; Experience with data analysis and experimental methods;

Preferred skills

Experience with conversational agents or AI-driven interactive systems; Experience with user studies or behavioral data collection, especially with children; Interest in social computing, play-based learning, or interaction design; Familiarity with multimodal interaction (speech, vision, or embodied agents);



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Shengdong Zhao

Professor

City University of Hong Kong

01. MMKG-Driven Exploration and Curation for a Martial Arts Heritage Archive

Multimodal Knowledge Graphs (MMKGs) provide the structured representation required for nuanced, real-world knowledge that traditional KG methods cannot effectively describe. This particularly applies to somatic traditions like martial arts, which cannot be reduced to purely cultural, technical, sportive, or performative manifestations, nor expressed through static arrays of textual or visual data.

This project leverages MMKGs to curate a multimodal archive in which 3D, motion capture, and video are the central data modalities, yet how to meaningfully present the graphs remains a challenge. Based on an established ontology and a preliminary dataset, the candidate will design and develop an engaging web experience integrating interactive visualisations for the MMKG and an MMKG-driven query system. As a further step, digital storytelling and curatorial innovation will be explored to transform archival data into extended heritage experiences.

Note: This project will be primarily supervised by Prof. Xiaoyu Zhang from City University of Hong Kong. Prof. Shendong Zhao will provide consultation only upon request and as necessary.

Required skills

Basic knowledge about data visualisation, Web design & development (Python & JavaScript) and an understanding of semantic web & knowledge graphs

Preferred skills

User Experience Design, experience with RDF and SPARQL, and enthusiasm for culture-related research



申请条件

1. 申请者须具有较高的英语听说、阅读及写作水平，如有相关成绩证明（雅思、托福、GRE等语言测试）请在简历文件中提供；
2. 目前为国内外高校计算机科学、电子工程、工业工程及相关专业的本科或硕士在读学生或毕业生，或具备同等专业能力者；
3. 有人机交互、人因工程、人工智能等方向的研究经历的申请者优先考虑，但非必需；
4. 本项目采用**滚动录取**，招满即止，请有意向的同学尽早提交申请。

报名方式

通过填写在线申请表报名，链接如下：

<https://wj.qq.com/s2/26569312/valy/>

(本申请表中需上传cv附件，建议在PC端和wifi环境下打开)

如有问题请咨询: hcixclub@gmail.com
(请在邮件主题上增加[HCIX]关键词)
或在评论区留言